

RITHIK BANERJEE

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EDUCATION

COLUMBIA UNIVERSITY — Fu Foundation School of Engineering

New York, NY

B.S. Computer Science, Minor in Applied Physics

Expected December 2026

GPA: **3.95/4.0**; Dean's List | Activities: IEEE, CU Informs, Wind Ensemble, Columbia Pops, Formula SAE Racing

Relevant Coursework: Analysis of Algorithms, Operating Systems, Fundamentals of Computer Systems, Artificial Intelligence, Quantum Computing, Reinforcement Learning, Neural Networks, Linear Algebra & Probability

WORK EXPERIENCE

C SPIRE (promoted from Systems Integration Intern, one of **top 5% performing** interns company-wide)

Ridgeland, MS

AI Solutions Engineer — AI Rapid Response Engineering Team

May 2026 – Present

- Architected and shipped a production voice-AI billing platform integrating WebEx, AWS, and distributed real-time LLM inference (~1.5 s latency), replacing proposed **\$3M+ vendor solution**, and establishing C Spire's standard architecture for future voice-AI products, saving **100+ development hours monthly**.
- Led a platform production security review for the distributed AI architecture spanning gRPC, AWS, private VPN infrastructure, and WebSockets; produced deployment audit documentation for cybersecurity approval.
- Designed, implemented, & deployed an LLM-based agent automating HR onboarding, saving **80+ HR staff hours/month**.
- Migrated an existing AI application onto a reusable production framework, extending adoption across teams and enabling additional AI products projected to save **50+ engineering/customer hours** each.
- Led technical development of a field-service automation platform integrating REST APIs, Kafka event streams, Redis, and SQL databases to sync status across ServiceNow and Jira for **100+ field technicians**, saving them **80+ hours/month**.
- Resolved **20 production issues** across six microservices using Grafana and distributed tracing; authored unit and integration test suites supporting **100+ production commits**.

COLUMBIA UNIVERSITY FUSION RESEARCH CENTER

New York, NY

Software Engineer — Scientific Computing | [rithikbanerjee314/simsomt](https://rithikbanerjee314.github.io/simsomt)

Feb 2025 – May 2026

- Developed 10+ Python optimization tools using SIMSOPT (an open-source fusion research library) to create **1000+ novel designs** for a next-generation clean-energy nuclear fusion reactor, improving target constraints by **30%**; optimized designs now guide physical coil construction on Columbia's High Beta Tokamak (HBT) fusion experiment.
- Automated high-performance computing (HPC) optimization workflows for large-scale simulation runs, adding engineering constraint validation to ensure optimized reactor designs remained physically manufacturable.
- Co-authored a **peer-reviewed arXiv preprint** on advanced nuclear fusion reactor design; presented findings at 2 symposia.

BROWN UNIVERSITY PRE-COLLEGE PROGRAM

Providence, RI

Teaching Assistant, "Computational Math and Data Visualization"

Jun 2024 – Aug 2024

- Delivered hands-on instruction in linear algebra, calculus, recursion, and NumPy-based programming to **20+ students**. Authored solution code for 8 programming assignments and graded 2 large-scale coding projects, delivering detailed individual feedback on algorithmic problem-solving and debugging.

PROJECTS

CRUCIBLE — High-Performance Chess Engine (chess.app)

[rithikbanerjee314/crucible](https://rithikbanerjee314.github.io/crucible), [rithikbanerjee314/chess](https://rithikbanerjee314.github.io/chess)

- Built a Universal Chess Interface (UCI)-compatible chess engine in C++ featuring a 14-technique alpha-beta search framework. Achieved a playing strength comparable to expert-level chess engines (**2468 Elo** against the industry-standard Stockfish benchmark) and verified move-generation correctness with a **119-million-position** performance test.

FISH — Distributed Multiplayer Game Server (fish-multiplayer.app)

[rithikbanerjee314/fish-multiplayer](https://rithikbanerjee314.github.io/fish-multiplayer)

- Architected and deployed an online multiplayer server for the card game “Fish” in Node.js using WebSockets, implementing authoritative server-side state management; validated with 4 automated integration tests, deployed on Render/Vercel.

FANTASY PREMIER LEAGUE TEAM MANAGER — Autonomous Decision Engine

[rithikbanerjee314/fpl-bot](https://rithikbanerjee314.github.io/fpl-bot)

- Built an autonomous Rust service that manages a Fantasy Premier League (online fantasy soccer) team by calling official APIs to automate player transfers, lineup selection, and game strategy. Designed independent optimization models and deployed continuously via GitHub Actions every six hours.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript/TypeScript, Rust, Scala, Julia, SQL, HTML/CSS

Systems & Backend: REST APIs, gRPC, WebSockets, Apache Kafka, Redis, Microservices Architecture, Distributed Systems, System Design, Multithreading, Linux

Cloud & DevOps: AWS, CI/CD (GitHub Actions), Unit & Integration Testing, Git/GitHub/GitLab, Grafana, Vercel, Render

Web & Data: React, Next.js, Node.js, SQL, SQLite, NumPy

AI/LLM: LLM Agent Development, Vercel AI Gateway, Claude API, Conversational & Voice AI